

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Civil Engineering) SEMESTER - 7 Winter 2025 (Supply.)

Course :Bachelor of Technology (Civil Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 7

Subject Code & Name: BTCVE705E - APPLICATIONS OF DRONE TECHNOLOGY

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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- Q1. Objective type questions. (Compulsory Question) 12
- 1 Which drone configuration offers the highest endurance for large-area mapping?
 - A) Quad-copter
 - B) Hexacopter
 - C) Fixed-wing
 - D) Tricopter
 - 2 Regulation of drones in India is handled by:
 - A) FAA
 - B) DGCA
 - C) NASA
 - D) ISRO
 - 3 In India, NPNT compliance is linked with:
 - A) Geofencing only
 - B) UIN and UAS Traffic Management
 - C) Drone battery registration
 - D) Pilot skill testing
 - 4 Ground Sampling Distance (GSD) is inversely proportional to:
 - A) Flying height
 - B) Sensor resolution
 - C) Focal length
 - D) Pixel size
 - 5 What does LiDAR stand for?
 - A) Light Detection and Ranging
 - B) Light Division and Radar
 - C) Linear Data Recording
 - D) Location Identifier and Radar
 - 6 LiDAR point density mainly depends on:
 - A) Laser color
 - B) Pulse repetition rate
 - C) Drone battery capacity
 - D) Wind direction

- 7 Drone-based structural inspection is especially useful for:
 - A) Identifying subsurface voids
 - B) Capturing surface anomalies like cracks and spalling
 - C) Concrete mix estimation
 - D) Cement hydration monitoring
- 8 For highway monitoring, temporal resolution indicates:
 - A) Ability to detect daily progress
 - B) Colour clarity of images
 - C) Drone flight noise
 - D) GPS antenna size
- 9 For accurate 3D reconstruction, the most important step is:
 - A) Colour balancing
 - B) Bundle adjustment
 - C) Contrast enhancement
 - D) Image watermarking
- 10 Radiometric correction is necessary to:
 - A) Normalize reflectance values
 - B) Adjust geometric distortions
 - C) Increase GPS accuracy
 - D) Correct wind drift
- 11 In transportation engineering, corridor mapping requires:
 - A) High forward overlap and narrow flight paths
 - B) Only one flight line
 - C) No GCPs
 - D) Random altitude variations
- 12 For forest fire assessment, the most suitable sensor is:
 - A) Multispectral
 - B) Thermal infrared
 - C) LiDAR
 - D) RGB

Q2. Solve the following.

- A) Explain the versatility of drones and their importance in modern civil engineering projects. 6
- B) Describe the regulatory framework governing drone usage in India and its significance for civil engineering professionals. 6

Q3. Solve the following.

- A) Compare drone-based surveying with traditional surveying methods in terms of accuracy, efficiency, and limitations. 6
- B) Describe LiDAR technology and its advantages in civil engineering data acquisition. 6

Q4. Solve Any Two of the following.

- A) Explain how drones improve construction site management and progress monitoring. 6
- B) Describe the role of drones in environmental monitoring and sustainability assessment. 6
- C) Discuss the significance of using drones in post-disaster assessment and emergency response. 6

- Q5. Solve Any Two of the following.
- A) Explain the process of converting raw drone imagery into an orthomosaic map. 6
 - B) Explain how GIS integration enhances drone-based mapping and analysis. 6
 - C) Discuss the importance of 3D modelling and point cloud analysis in civil engineering projects. 6
- Q6. Solve Any Two of the following.
- A) Describe the use of drones in transportation engineering for road and highway monitoring. 6
 - B) Discuss the role of drones in infrastructure inspection, including bridges, buildings, and dams. 6
 - C) Discuss the application of drones in post-disaster damage assessment and how they enhance recovery operations. 6

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